T-TEST — Complete Study Guide

What is a T-Test?

A T-Test is used exactly like a Z-Test — to check if a sample mean is significantly different from a population mean. The key difference is that **T-Test is used when the population standard deviation (σ) is UNKNOWN**, and you only have the **sample standard deviation (s)**. This is the most common real-world scenario!

Z-Test vs T-Test — The Key Difference

	Z-Test	T-Test
When to use	σ (population std) KNOWN	σ UNKNOWN — use sample std (s)
Real world?	Rare	Very Common
Sample size	n > 30 preferred	Works for any n
Table used	Z-Table	T-Table (needs df)
Extra input	None	Degrees of Freedom: df = n-1
Formula diff	Uses σ	Uses s instead of σ

The T-Test Formula

t = (x■ – μ) / (s / √n)

Symbol	Meaning	Example (IQ Problem)
x■	Sample Mean	140
μ	Population Mean (claim)	100
s	Sample Std Deviation	20
n	Sample Size	30
df	Degrees of Freedom = n-1	29
t	Calculated T score	10.95

What are Degrees of Freedom (df)?

df = n – 1. Think of it this way — if you have 30 values and you know the mean, only 29 values are truly "free" to vary (the last one is fixed). **The smaller the df, the wider the T-distribution tails** — because with fewer data points, there is more uncertainty. As df increases, the T-distribution looks more and more like the Z-distribution.

Degrees of Freedom (df)	Critical T (α=0.05, two-tail)	Comparison
5	2.571	Much wider than Z

10	2.228	Still wider than Z
20	2.086	Getting closer
29	2.045	Our problem!
30	2.042	Very close to Z
∞	1.960	Same as Z-table!

Key Insight: When df is very large ($n > 120$), T-table and Z-table give almost the same values. That's why Z-test and T-test converge for large samples!

PROBLEM — IQ Medication Example (Two-Tail T-Test)

In the population, the average IQ is **100**. Researchers want to test a new medication to see if it has either a positive or negative effect on intelligence. A sample of **30 participants** who took the medication has a mean IQ of **140** with a standard deviation of **20**. Did the medication affect intelligence? (CI = 95%, $\alpha = 0.05$)

Given Values

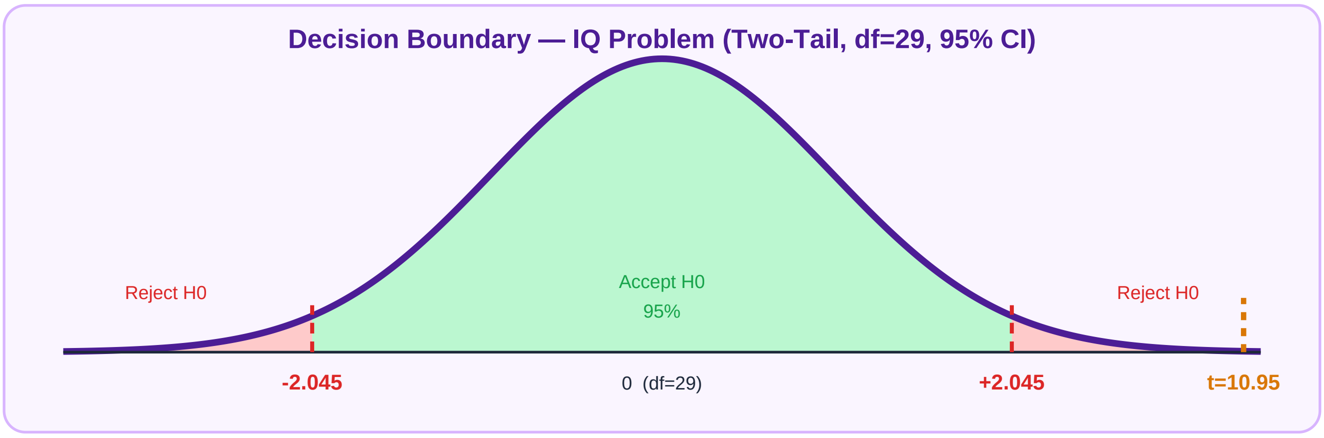
μ (Pop Mean)	x■ (Sample Mean)	s (Sample Std)	n	df = n-1	α
100	140	20	30	29	0.05

Why T-Test? (Not Z-Test!)

✓ Population σ is **NOT given** — only sample std (s=20) is given
✓ We use **s** in the formula instead of σ
✓ We need **df** to look up T-table instead of Z-table

Step-by-Step Solution

1	**State Hypotheses**	H0: μ = 100 (m H1: μ ≠ 100 (m → TWO-TAIL T
2	**Calculate Degrees of Freedom**	df = n – 1 = 30 This is needed
3	**Find Critical T Value from T-Table**	Look up T-table T-critical = ±2.0 Rule: Reject H0
4	**Calculate T Score**	t = (x■ – μ) / (s = (140 – 100) / = 40 / (20/5.47 = 40 / 3.65 = 10.95
5	**Compare & Make Decision**	t = 10.95 > t_cr 10.95 falls WA p-value << 0.05



FINAL CONCLUSION: t = 10.95 is far beyond the critical value of ±2.045

We **REJECT the Null Hypothesis**. The medication DID significantly affect intelligence.

The average IQ of 140 is statistically very different from the population mean of 100.

QUICK REFERENCE CHEAT SHEET

How to Read the T-Table

Unlike the Z-table (which uses area), the T-table uses α **and df** directly.

Step	What to do	Our Example
1	Find df = n-1	df = 30-1 = 29
2	Find your α (two-tail or one-tail)	α = 0.05, two-tail
3	Find the row for df=29	Row 29
4	Find the column for your α	Column 0.05 (two-tail)
5	Read the intersection value	t_critical = 2.045

Complete T-Test Process

STEP 1	**Identify Inputs** μ, s, n, x■, CI or α
STEP 2	**Check conditions** σ unknown? → T-test. σ known? → Z-test
STEP 3	**Set Hypotheses** H0 = claim. H1 = what you want to prove
STEP 4	**Identify tail type** H1 uses ≠ → Two-tail | < → Left | > → Right
STEP 5	**Calculate df** df = n – 1
STEP 6	**Find t_critical** Look up T-table using df and α
STEP 7	**Calculate t score** t = (x■ – μ) / (s/√n)
STEP 8	**Make Decision** |t| > t_critical → Reject H0 |t| ≤ t_critical → Fail to Reject H0

Z-Test vs T-Test — Full Comparison

	Z-Test	T-Test
σ known?	YES	NO
Uses	σ in formula	s in formula

Extra calc	None	df = n-1
Lookup table	Z-table	T-table (with df)
Critical value (95% CI, two-tail)	±1.96	±2.045 (df=29)
Real world use	Rare	Very Common
Conclusion logic	Same	Same

T-Test: σ unknown → use s → need df = n-1 → T-table | Everything else is same as Z-Test!